Poetry GP: Learning Preference with Gaussian Processes and HPC

CS 2050 - High Performance Computing

Spring 2026

Poetry GP: Learning Preference

Gaussian Processes, Active Learning, and HPC

Goal: Learn reader's taste, recommend poems strategically

- 85,000 poems $\rightarrow$ 384-dim embeddings
- User rates poems $\rightarrow$ Infer preference function
- Balance **exploit** (recommend favorites) vs **explore** (ask questions)

HPC Challenge: Exact GP scales as $O(m^3)$ fit, $O(nm^2)$ scoring

Problem Motivation

Authentic ML application:

- High-dimensional data ($d = 384$)
- Large candidate pool ($n = 85{,}000$)
- Interactive learning (ratings accumulate)
- Uncertainty quantification (know confidence)

Computational bottlenecks:

- **Fit**: $O(m^3)$ Cholesky factorization
- **Score**: $O(nm^2)$ variance computation
- Real performance challenges motivate HPC solutions

Pedagogical value: Theory meets practice

The Learning Loop

```
Rate poem → Update posterior → Score candidates → Choose next poem
  (input)      (O(m³) fit)          (O(nm²) score)     (acquisition)
```

Given:

- Rated: $\{(x_i, y_i)\}_{i=1}^{m}$ where $x_i \in \mathbb{R}^{384}$
- Candidates: $\{x_j\}_{j=1}^{85000}$

Compute:

- Posterior mean $\mu(x)$ (expected preference)
- Posterior variance $\sigma^2(x)$ (uncertainty)

Ridge Regression = Bayesian Linear Regression

Linear model:

$$y = X\beta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_n^2 I)$$

Gaussian prior (ridge penalty):

$$\beta \sim \mathcal{N}(0, \tau^2 I)$$

MAP estimate:

$$\hat{\beta} = (X^\top X + \lambda I_p)^{-1} X^\top y$$

Prediction:

$$\hat{y}_* = x_*^\top \hat{\beta}$$

Interpretation: Ridge shrinkage = Bayesian prior

Dual Problem: Primal vs Dual

Primal form ($p \times p$ system):

$$\hat{y}_* = x_*^\top (X^\top X + \lambda I_p)^{-1} X^\top y$$

Dual form ($m \times m$ system):

$$\hat{y}_* = x_*^\top X^\top (XX^\top + \lambda I_m)^{-1} y$$

Define Gram matrix $K = XX^\top$ and $k_* = Xx_*$:

$$\hat{y}_* = k_*^\top (K + \lambda I)^{-1} y$$

Key insight: All predictions use inner products $K_{ij} = x_i^\top x_j$

When $m < p$: Dual is cheaper

From Inner Products to Kernels

Linear kernel:

$$K_{ij} = x_i^\top x_j$$

Replace with general kernel:

$$K_{ij} = k(x_i, x_j)$$

RBF kernel (our choice):

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right)$$

Prediction with kernel:

$$\hat{y}_* = k_*^\top (K + \lambda I)^{-1} y$$

Interpretation: RBF $\rightarrow$ nearby poems have similar ratings

Gaussian Process Prior

GP as multivariate Gaussian over function values:

For two points x_1, x_2:

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} k(x_1, x_1) & k(x_1, x_2) \\ k(x_2, x_1) & k(x_2, x_2) \end{bmatrix} \right)$$

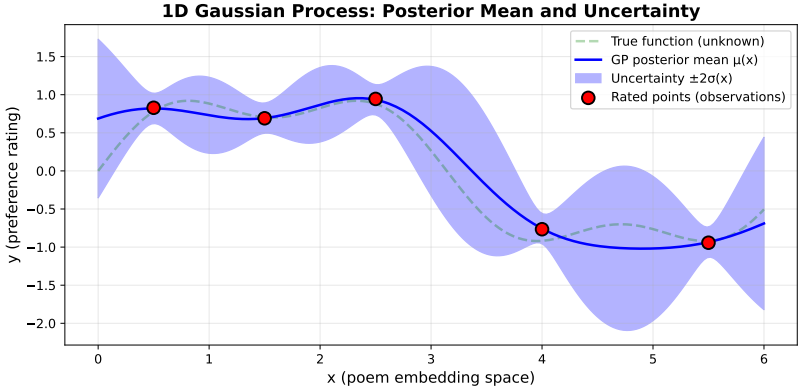
Generalize to m rated points:

$$f_r \sim \mathcal{N}(0, K_{rr})$$

where $(K_{rr})_{ij} = k(x_i, x_j)$

Key idea: Kernel defines covariance between function values

1D GP Example



Observations: Sparse rated points (red dots)

Posterior: Mean $\mu(x)$ (blue line), uncertainty $\pm 2\sigma(x)$ (shaded)

Uncertainty shrinks near observations, **grows** in unexplored regions

GP Posterior Update

Joint distribution (rated + new point):

$$\begin{bmatrix} y_r \\ f_* \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} K_{rr} + \sigma_n^2 I & k_{r*} \\ k_{*r} & k_{**} \end{bmatrix} \right)$$

Posterior mean:

$$\mu(x_*) = k_{*r}^\top (K_{rr} + \sigma_n^2 I)^{-1} y_r$$

Posterior variance:

$$\sigma^2(x_*) = k(x_*, x_*) - k_{*r}^\top (K_{rr} + \sigma_n^2 I)^{-1} k_{r*}$$

This is what we need to compute

Computational Bottleneck: Fit

What we compute (define $K = K_{rr} + \sigma_n^2 I$):

1. **Cholesky factorization**: $K = LL^\top$
2. **Solve for weights**: $\alpha = K^{-1}y$

Complexity:

- Kernel assembly: $O(m^2 d)$
- **Cholesky**: $O(m^3) \leftarrow$ **bottleneck**
- Solve: $O(m^2)$

Scaling variable: $m =$ rated poems

As m **grows**: Cubic scaling!

Computational Bottleneck: Scoring

For n candidates, compute:

Posterior mean (efficient):

$$\mu_q = K_{qr}\alpha \qquad \text{Cost: } O(nmd + nm)$$

Posterior variance (expensive):

$$\sigma_q^2 = \text{diag}(K_{qq}) - \text{diag}(K_{qr}K^{-1}K_{rq})$$

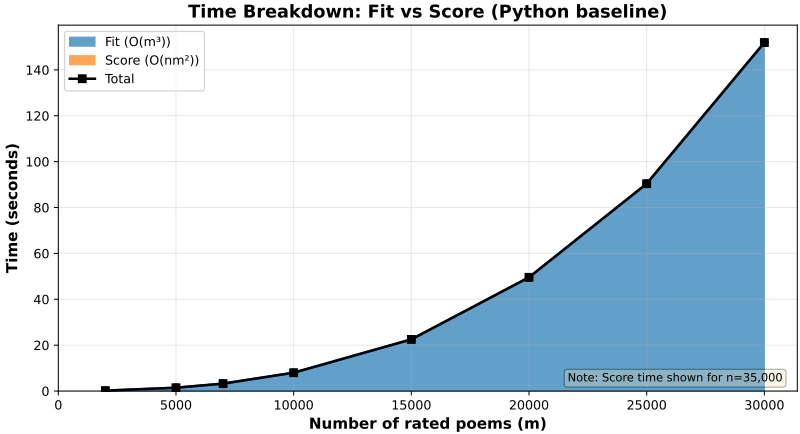
Rewrite as $v = L^{-1}K_{rq}$:

$$\sigma_q^2 = \text{diag}(K_{qq}) - \text{diag}(v^\top v)$$

Cost: $O(nm^2) \leftarrow$ **bottleneck for exploration**

Key variables: $n = 85{,}000$ candidates, $m =$ rated poems

Time Breakdown: Fit vs Score



Fit dominates for large m (cubic scaling), **score dominates** for large n with variance

Hyperparameter Optimization

Kernel hyperparameters: $\theta = (\ell, \sigma_f, \sigma_n)$

- Length scale ℓ: How far correlations extend
- Signal variance σ_f^2: Prior belief about rating scale
- Noise σ_n^2: Measurement noise

Optimization: Maximize marginal likelihood

$$\log p(y \mid X, \theta) = -\frac{1}{2} y^\top K^{-1} y - \frac{1}{2} \log |K| - \frac{m}{2} \log(2\pi)$$

Method: L-BFGS with analytic gradients

Cost: $O(m^3)$ per evaluation (same as fit)

When: Before interactive session or periodically during use

Active Learning: Acquisition Functions

Exploitation (recommend favorites):

- max_mean: $\arg\max \mu(x)$ — greedy, fast ($O(nm)$)
- UCB: $\arg\max[\mu(x) + \beta\sigma(x)]$ — balance ($O(nm^2)$)
- thompson: Sample $f \sim \mathcal{N}(\mu, \Sigma)$ — Bayesian ($O(nm^2)$)

Exploration (ask informative questions):

- max_variance: $\arg\max \sigma^2(x)$ — information-optimal ($O(nm^2)$)
- expected_improvement: Classic BO ($O(nm^2)$)
- spatial_variance: Spatially diverse ($O(n^2)$, slow)

Computational tradeoff: Exploration requires variance $\rightarrow O(nm^2)$

Implementation: Backend Abstraction

```
result = run_blocked_step(
    embeddings, rated_indices, ratings,
    fit_backend="auto",       # python, native_lapack, scalapack
    score_backend="auto",     # python, native_lapack, gpu
)
```

Fit backends ($O(m^3)$):

- python: NumPy/SciPy (baseline)
- native_lapack: PyBind11 + LAPACK (zero overhead)
- scalapack: MPI distributed (block-cyclic layout)

Score backends ($O(nm^2)$):

- python: NumPy + BLAS
- native_lapack: Multi-threaded BLAS
- gpu: CuPy/CUDA

Fit Backend: Python

```
K = rbf_kernel(X_rated) + noise * np.eye(m)
L = scipy.linalg.cholesky(K, lower=True)
alpha = scipy.linalg.cho_solve((L, True), y)
```

Characteristics:

- Pure Python + NumPy
- ~5ms overhead (Python calls, allocations)

When to use: Small problems ($m < 5{,}000$), prototyping

Fit Backend: PyBind11 + LAPACK

```
// Direct LAPACK, no Python overhead
dpotrf_(&uplo, &m, K.data(), &m, &info);  // Cholesky
dtrsv_(...);  // Triangular solve
```

Characteristics:

- Eliminate Python overhead
- Shared memory (single node)
- ~0.1ms overhead (nearly zero)

When to use: $m < 5{,}000$, single-node sufficient

Fit Backend: ScaLAPACK (Distributed)

Process grid ($P \times Q$):

```
2x2 grid:
+-----+-----+
| P00 | P01 |
+-----+-----+
| P10 | P11 |
+-----+-----+
```

Block-cyclic distribution (block size b):

- Matrix divided into $b \times b$ blocks
- Distributed across processes in cyclic pattern
- Enables parallel Cholesky factorization

Key optimization (Milestone 1B): Broadcast features (~30MB) instead of scatter matrix (~800MB) $\rightarrow$ 20-40$\times$ assembly speedup

Benchmark: Fit Scaling (Actual Data)

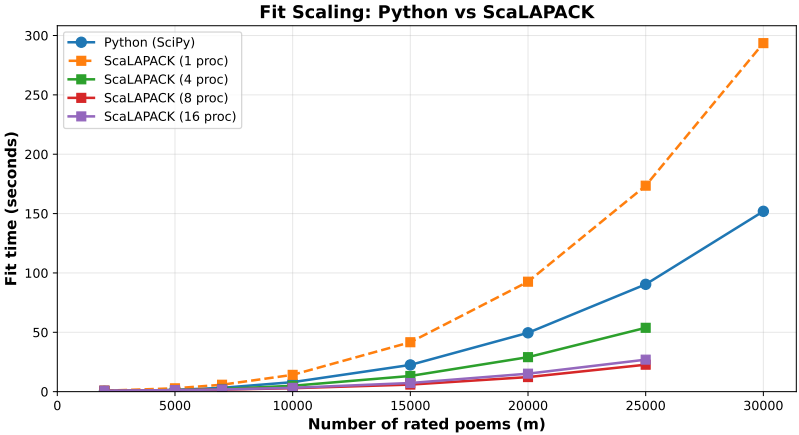
Python (SciPy) vs ScaLAPACK (varying processes):

m	Python	ScaLAPACK (1 proc)	ScaLAPACK (4 proc)	ScaLAPACK (8 proc)	ScaLAPACK (16 proc)
2,000	0.18s	0.77s	0.64s	0.69s	0.86s
7,000	3.24s	5.51s	2.13s	**1.58s**	1.83s
15,000	22.51s	38.00s	12.66s	**5.52s**	7.03s
20,000	49.57s	84.79s	27.44s	**11.56s**	14.88s

Key findings:

- Crossover: $m \approx 5{,}000 - 7{,}000$
- Best: 8 processes (4.3× faster at $m = 20{,}000$)
- 16 processes: Diminishing returns (overhead)

Benchmark: Fit Scaling (Visual)



Observations: Python fast for small m, ScaLAPACK wins at large m, 8 processes optimal

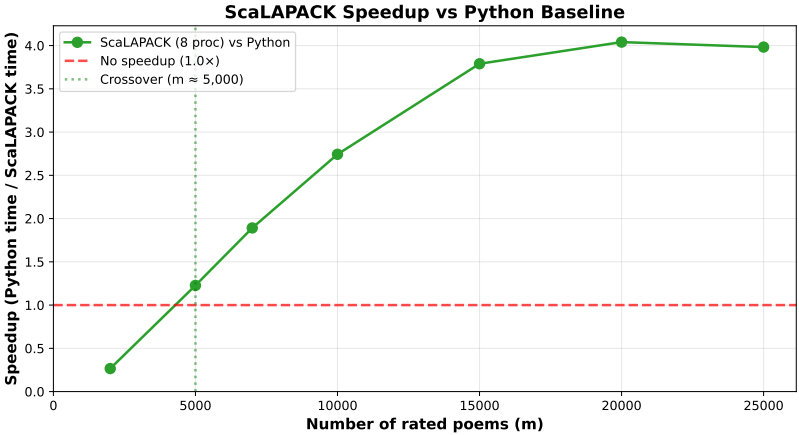
Benchmark: ScaLAPACK Speedup

Speedup vs Python (ScaLAPACK with 8 processes):

m	Python time	ScaLAPACK time	Speedup
2,000	0.18s	0.69s	**0.26×** (slower!)
5,000	1.45s	1.12s	1.3×
7,000	3.24s	1.58s	2.1×
10,000	8.00s	2.59s	3.1×
15,000	22.51s	5.52s	4.1×
20,000	49.57s	11.56s	**4.3×**

Observation: Overhead dominates for $m < 5{,}000$, then compute wins

Benchmark: Speedup (Visual)



Crossover at $m \approx 5{,}000$: Below = overhead dominates, above = parallelism wins

Benchmark: Why 16 Processes Slower?

ScaLAPACK at $m = 20{,}000$:

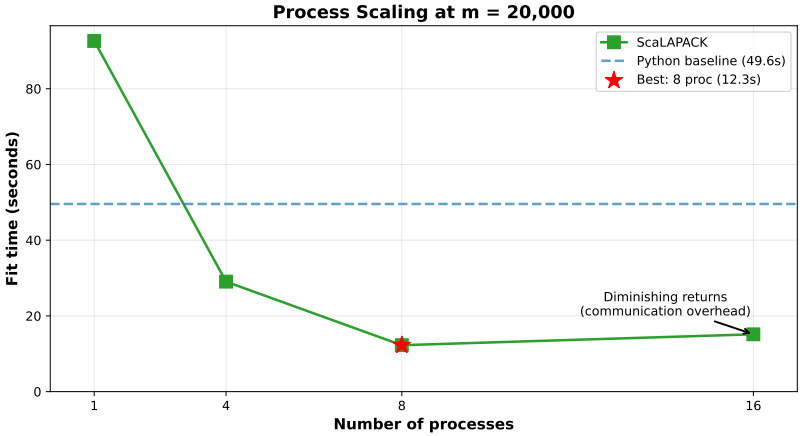
- 1 process: 84.79s (sequential overhead)
- 4 processes: 27.44s
- 8 processes: **11.56s** (best)
- 16 processes: 14.88s (worse than 8!)

Interpretation:

- Communication overhead grows with process count
- Problem size not large enough for 16 processes
- Amdahl's law: Fixed overhead limits speedup

Lesson: More processes $\neq$ better performance

Benchmark: Process Scaling (Visual)



Amdahl's law in action: 8 processes optimal, 16 shows diminishing returns

War Stories: What Worked

[WORKS] Distributed kernel assembly:

- Broadcast features (30MB), not matrix (800MB)
- 20-40$\times$ assembly speedup

[WORKS] PyBind11 for single-node:

- Eliminate Python overhead
- Best for $m < 5{,}000$

[WORKS] Lazy variance evaluation:

- Skip $O(nm^2)$ when not needed
- max_mean is 85$\times$ faster than max_variance

War Stories: What Didn't Work (At First)

[FAILED] ScaLAPACK performance mystery:

- Pedagogical benchmarks 21-146× slower than expected
- Root cause: Missing `OMP_NUM_THREADS=1` (BLAS threading conflict)
- Root cause: Wrong MPI launcher (`srun` vs `mpirun`)
- Fix: Environment variable hygiene

[FAILED] PyBind11 on compute nodes:

- Build fails: "Python.h not found"
- Lesson: PyBind11 for shared memory only

[FAILED] Concurrent build conflicts:

- Multiple jobs clobbering `native/build/`
- Fix: Job-specific build dirs

Key lesson: Environment details matter in HPC!

Current Performance

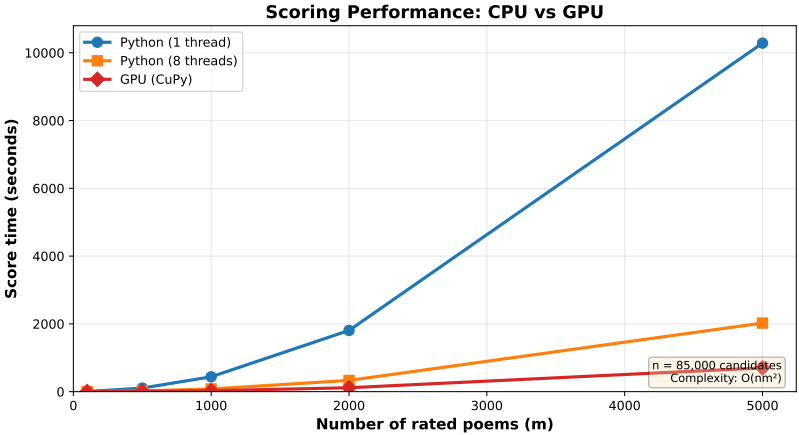
Interactive session ($m = 1{,}000$, $n = 85{,}000$):

Phase	Backend	Time
Fit	native_lapack (PyBind11)	30ms
Score	GPU (CuPy)	760ms
Select	max_variance	1ms
Total		**791ms**

User experience: Sub-second recommendations with uncertainty

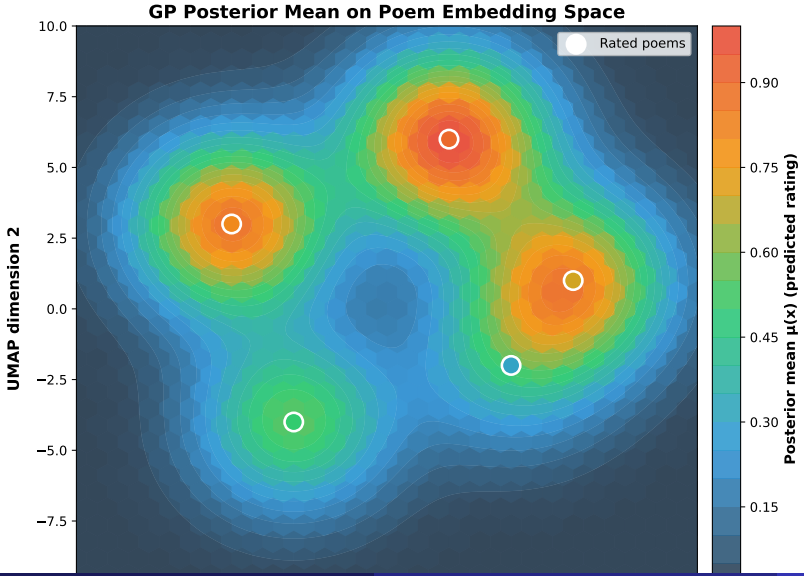
Backend choices: PyBind11 for fit (fast), GPU for score (variance needed)

Benchmark: Score Backend Comparison



GPU wins for scoring: 15× faster than single-threaded Python at $m = 5{,}000$

Posterior Visualization



Try It Yourself!

Quick start:

```
# 1. Setup environment
bash scripts/bootstrap_env.sh
source scripts/activate_env.sh

# 2. Get shared data (or build from scratch)
bash scripts/setup_shared_data.sh

# 3. Run interactive CLI
python scripts/app/interactive_cli.py
```

Features: - Rate poems → GP updates in real-time - Exploit (e): Get recommendations (UCB, Thompson) - Explore (x): Ask informative questions (max_variance) - Visualize: Posterior heatmaps

Repository: `github.com/cgumb/poetry` (or share link)

Questions?

Future Work

Model: Full Bayesian hyperparameters (MCMC), learned features (NN + GP), alternative kernels

Active learning: RL for acquisition, batch queries, contextual bandits

Computational: Sparse GPs (inducing points), GPU kernel assembly, distributed scoring